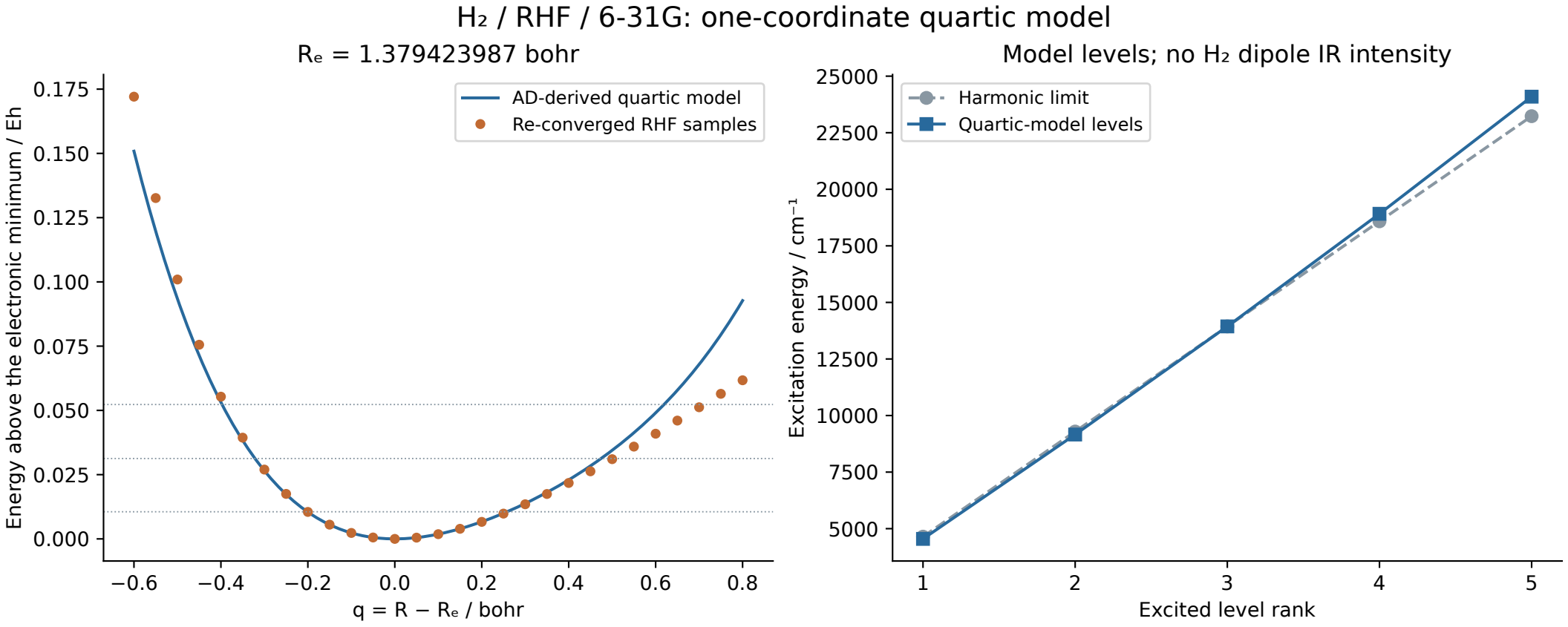
H₂ / RHF / 6-31G: one-coordinate quartic model
Rₑ = 1.379423987 bohr
AD-derived quartic model
Re-converged RHF samples
Energy above the electronic minimum / Eh
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
q = R − Rₑ / bohr
Model levels; no H₂ dipole IR intensity
Harmonic limit
Quartic-model levels
Excitation energy / cm⁻¹
25000
22500
20000
17500
15000
12500
10000
7500
5000
1
2
3
4
5
Excited level rank